

```
In [1]: from textblob import TextBlob
import nltk
nltk.download('punkt')
nltk.download('stopwords') # For removing common words
nltk.download('averaged_perceptron_tagger') # For Part-of-Speech tagging (us
nltk.download('wordnet') # For lemmatization
! python -m textblob.download_corpora
# Sample reviews with associated sentiment (as provided)
```

```
[nltk_data] Downloading package punkt to /Users/mathis/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data] /Users/mathis/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /Users/mathis/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
[nltk_data] Downloading package wordnet to /Users/mathis/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
zsh:1: command not found: python
```

```
In [2]: import pandas as pd
import re
from collections import Counter

import matplotlib.pyplot as plt
import seaborn as sns
import nltk
from textblob import TextBlob, Word
from wordcloud import WordCloud
from nltk.corpus import stopwords

df = pd.read_csv("../datasets/amazon.csv")
df
```

Out [2]:

	reviewText	Positive
0	This is a one of the best apps acording to a b...	1
1	This is a pretty good version of the game for ...	1
2	this is a really cool game. there are a bunch ...	1
3	This is a silly game and can be frustrating, b...	1
4	This is a terrific game on any pad. Hrs of fun...	1
...	...	...
19995	this app is fricken stupid.it froze on the kin...	0
19996	Please add me!!!! I need neighbors! Ginger101...	1
19997	love it! this game. is awesome. wish it had m...	1
19998	I love love love this app on my side of fashio...	1
19999	This game is a rip off. Here is a list of thin...	0

20000 rows x 2 columns

1. Can you perform a sentiment analysis on the collection of product reviews, classify each one as 'positive', 'negative', or 'neutral', and then generate a bar chart to visualize the distribution of these sentiments?

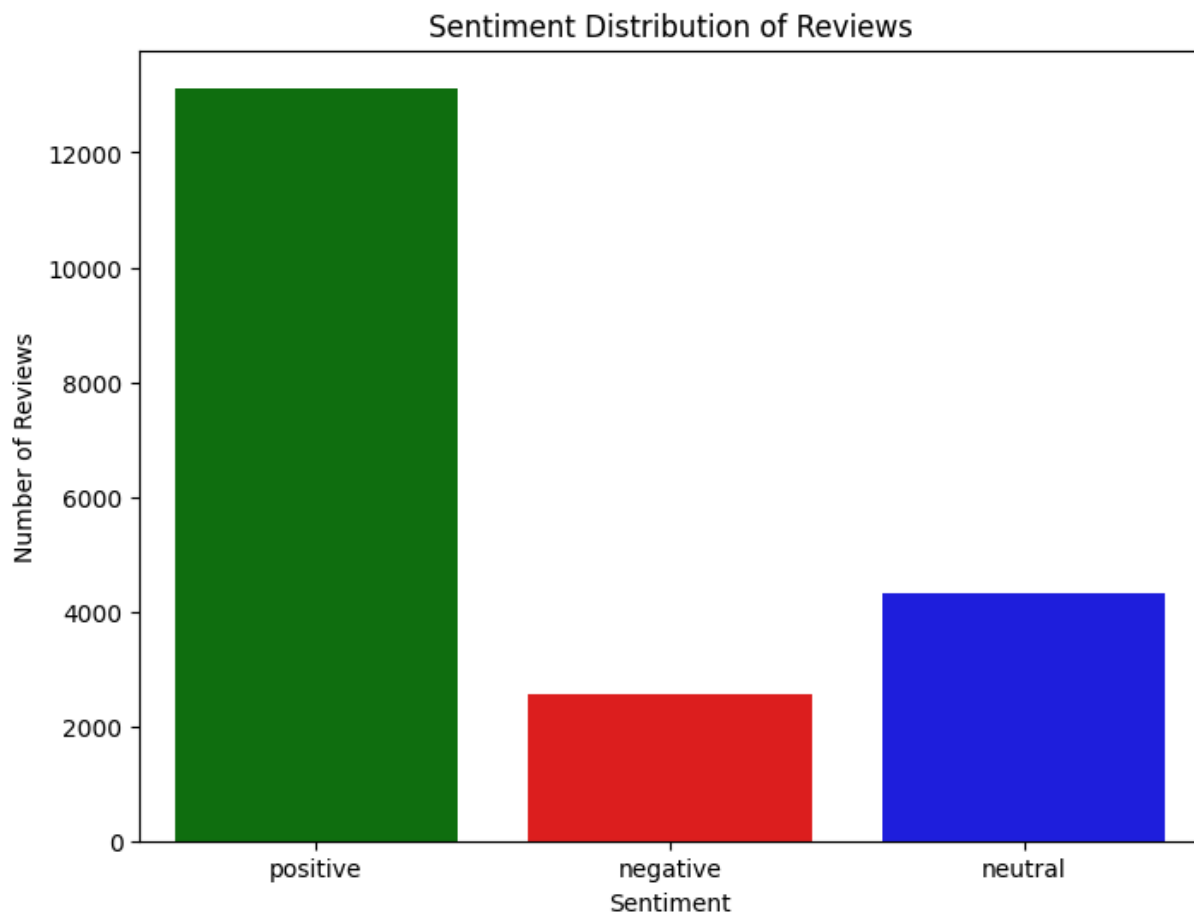
```
In [3]: text_col = 'reviewText' if 'reviewText' in df.columns else df.select_dtypes(
texts = df[text_col].fillna('').astype(str)
texts = texts[texts.str.strip() != '']

polarities = texts.apply(lambda x: TextBlob(x).sentiment.polarity)
sentiments = polarities.apply(lambda p: 'positive' if p > 0.1 else ('negative' if p < -0.1 else 'neutral'))

sentiment_counter = Counter(sentiments)

plt.figure(figsize=(8, 6))
# Updated seaborn API to avoid futures warnings
sns.barplot(x=list(sentiment_counter.keys()), y=list(sentiment_counter.values))
plt.title('Sentiment Distribution of Reviews')
plt.xlabel('Sentiment')
plt.ylabel('Number of Reviews')
plt.show()

print('Text column used:', text_col)
print('Reviews analyzed:', len(texts))
print('Sentiment counts:', sentiment_counter)
```



Text column used: reviewText

Reviews analyzed: 20000

Sentiment counts: Counter({'positive': 13127, 'neutral': 4324, 'negative': 2549})

- convert the text to lowercase, remove punctuation and numbers, filter out common stopwords, and then lemmatize the remaining words using their appropriate part-of-speech tags.

```
In [4]: nltk.download('punkt', quiet=True)
nltk.download('stopwords', quiet=True)
nltk.download('averaged_perceptron_tagger_eng', quiet=True)
nltk.download('wordnet', quiet=True)

stop_words = set(stopwords.words('english'))

def get_wordnet_pos(tag):
    if tag.startswith('J'):
        return 'a'
    if tag.startswith('V'):
        return 'v'
    if tag.startswith('N'):
        return 'n'
    if tag.startswith('R'):
        return 'r'
    return 'n'
```

```

all_lemmas = []
all_pos_tags = []

for text in texts:
    clean_text = re.sub(r'^a-zA-Z\s', ' ', text.lower())
    clean_text = re.sub(r'\s+', ' ', clean_text).strip()
    if clean_text == '':
        continue

    for word, tag in TextBlob(clean_text).tags:
        if word in stop_words or len(word) <= 1:
            continue
        lemma = Word(word).lemmatize(get_wordnet_pos(tag))
        all_lemmas.append(lemma)
        all_pos_tags.append(tag)

print('Number of lemmatized tokens:', len(all_lemmas))

```

Number of lemmatized tokens: 328184

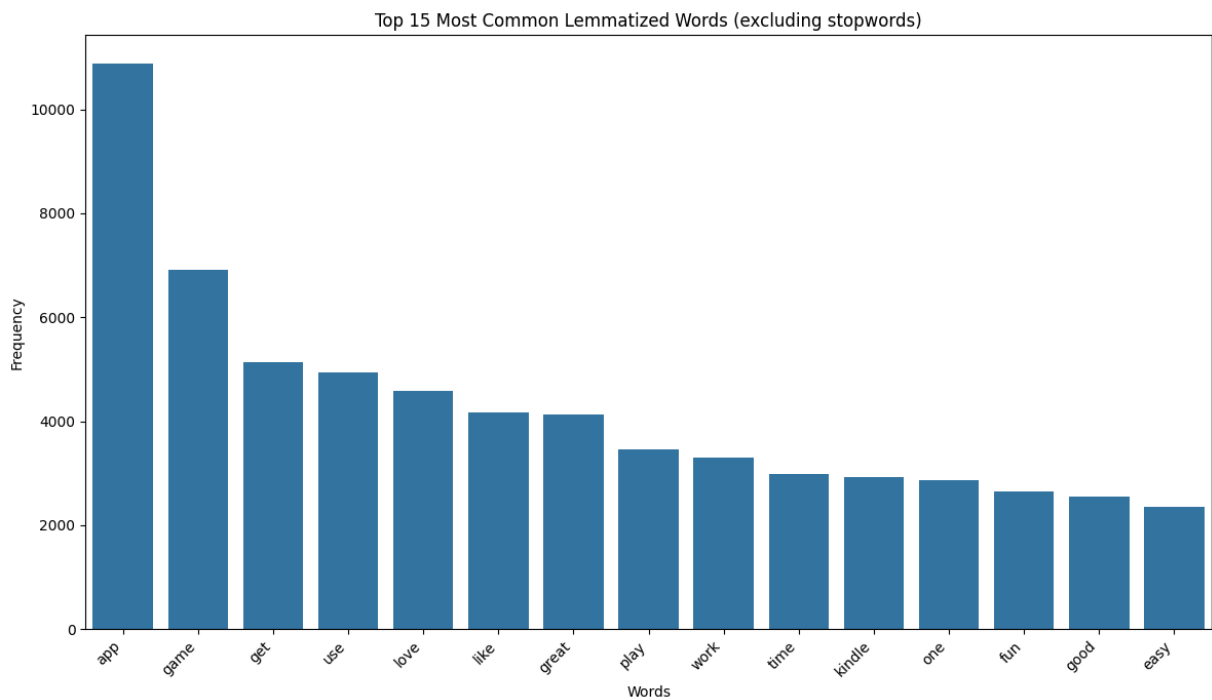
3. after the text has been fully preprocessed and lemmatized, could you calculate the frequency of each unique word and then display a bar graph showing the top 15 most common words?

```

In [5]: word_counts = Counter(all_lemmas)
top_15 = word_counts.most_common(15)

if len(top_15) > 0:
    words, counts = zip(*top_15)
    plt.figure(figsize=(12, 7))
    sns.barplot(x=list(words), y=list(counts))
    plt.title('Top 15 Most Common Lemmatized Words (excluding stopwords)')
    plt.xlabel('Words')
    plt.ylabel('Frequency')
    plt.xticks(rotation=45, ha='right')
    plt.tight_layout()
    plt.show()
else:
    print('No words found after preprocessing.')

```



4. generate a word cloud based on the entire corpus of lemmatized review words to provide a visual summary of the most prominent terms.

```
In [6]: corpus = ' '.join(all_lemmas)
if corpus.strip() != '':
    wc = WordCloud(width=800, height=400, background_color='white').generate
    plt.figure(figsize=(10, 8))
    plt.imshow(wc, interpolation='bilinear')
    plt.axis('off')
    plt.title('Visual Resume of Reviews (Word Cloud)')
    plt.show()
else:
    print('No words available for word cloud.')
```

[illegible]

5. Visualize the frequency of nouns, verbs, adjectives in this text.

```
In [7]: pos_counts = Counter({'nouns': 0, 'verbs': 0, 'adjectives': 0})
for tag in all_pos_tags:
    if tag.startswith('N'):
        pos_counts['nouns'] += 1
    elif tag.startswith('V'):
        pos_counts['verbs'] += 1
    elif tag.startswith('J'):
        pos_counts['adjectives'] += 1

plt.figure(figsize=(8, 6))
sns.barplot(x=['nouns', 'verbs', 'adjectives'], y=[pos_counts['nouns'], pos_
plt.title('Frequency of Nouns, Verbs, and Adjectives')
plt.xlabel('POS Category')
plt.ylabel('Count')
plt.show()
```

